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Original Research Article

Grid-cell based assessment of soil erosion potential for identification of critical erosion prone areas using USLE, GIS and remote sensing: A case study in the Kapgari watershed, India

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ABSTRACT

Estimation of soil erosion is of paramount importance due to its serious environmental and societal concern. Soil erosion would have impact on fertility of agricultural land and quality of water. The major objective of this study was to investigate the spatial heterogeneity of annual soil erosion on the grid-cell basis in a small agricultural watershed of eastern India. The study watershed has a drainage area of 973 ha and is subdivided into three sub-watersheds namely: KGSW1, KGSW2 and KGSW3, based on the land topography and drainage network. Average annual soil erosion was estimated on 100 m × 100 m grid-cells by integrating universal soil loss equation (USLE) model with GIS for subsequent identification of critical erosion prone areas. It was found that 82.63% area of the total watershed falls under slight-erosion-class (0–5 t-ha^{-1}-yr^{-1}), 6.87% area lies under the moderate-erosion-class (5–10 t-ha^{-1}-yr^{-1}), 5.96% area is under high-erosion-class (10–20 t-ha^{-1}-yr^{-1}), 3.3% area of watershed lies under the very-high-erosion-class (20–40 t-ha^{-1}-yr^{-1}) and 1.24% area falls under "severe-erosion-class" (40–80 t-ha^{-1}-yr^{-1}). The study revealed that the sub-watershed KGSW3 is critical due to the presence of the highest number of critical erosion prone grid-cells. The sediment delivery ratio (SDR) was also estimated to analyze the contribution of sediment yield at the sub-watershed level. Lowest SDR for the whole watershed as compared to sub-watersheds indicates that most of the eroded soil got deposited in rice crop check-basins before reaching the outlet. The reported results can be used for prioritizing critical erosion prone areas and for determining appropriate soil erosion prevention and control measures.
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1. Introduction

The soil erosion has increased during 20th century, and has become a worldwide issue of significant environmental and societal concern (Angima, Stott, O'Neill, Ong, & Weesies, 2003). India has a total geographical area of 329 Mha out of which 175 Mha (53%) is suffering from the land degradation problem. It has been assessed that nearly 5334 Mt soil erosion occurs yearly in India, due to various reasons of which about 10% settle down in the reservoirs, and 29% reaches the sea and significantly reduce the storage capacity (Narayana & Babu, 1983). The prevention of soil erosion and sediment deposition are important due to their direct impact on fertility of agricultural land and quality of water. About 85% of land degradation globally is due to by soil erosion, causing decline in crop yield up to 17% (Oldeman, Hakkeling, & Sombroek, 1990). Reduction in fertility of agricultural land as a consequence of soil erosion increases the expenses on fertilizers initially but afterward may lead to land abandonment (Pimentel, Harvey, Resosudarmo, & Sinclair, 1995). On the other hand the sedimentation at downstream area decreases the storage capacity of streams and retention ponds, which increases the possibility of flooding and reduces the designed life of water resources structures (Boardman, Ligneau, de Roo, & Vandaele, 1994; Verstraeten & Poesen, 1999). Deterioration of the drainage systems in agricultural watersheds has severe effect on soil erosion, which increases the water logging and salinity in agricultural fields (Valipour, 2014). The sediment is also a pollutant due to agro-chemicals adsorption, which can raise the nitrogen and phosphorus levels in water bodies and result in eutrophication (Sibbesen, 1995; Steegen et al., 2001).

Soil erosion may be more severe in near future due climatic change across various parts of the world (Amore, Modica, Nearing, & Santoro, 2004). The soil erosion will aggravate further with increasing population pressure, over-utilization of natural resources, faulty land and water management practice (Jena et al., 2015). There is a need for soil conservation to reverse the process of land

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abandonment and enhancement in agricultural production to ensure food security and sustainability. Hence, there is a necessity for identifying critical erosion prone areas at watershed scale to provide pre-requisite information for effective watershed management, including soil conservation strategies.

Appropriate watershed management plans for sustainable development need reliable long-term soil loss information at various parts of the watershed. Thus, many hydrological models such as: empirical, lumped conceptual and physically based models are being utilized for decades for assessment of soil erosion potential (de Vente & Poesen, 2005; Lal, 2001). Investigation in the past revealed that empirical models are relatively simple, robust in structure and have significant importance in assessment of soil erosion potential under scarcity of field data (Zhu, 2015).

The universal soil loss equation (USLE), is one of the most popular empirical models (Wischmeier & Smith, 1978) to estimate the long-term average annual rate of soil loss from small field having an average length of 22 m, a field slope of 9% based on rainfall pattern, soil type, topography, cropping system and management practices. In the last few decades, remote sensing (RS) technology and geographic information system (GIS) were used to precisely estimate the soil erosion precisely on the watershed and basin scale rather at a field scale (Chen, Niu, Li, Zhang, & Du, 2012; Cohen, 1960; Gelagay & Minale, 2016; Kouli, Soupios, & Vallianatos, 2009; Mhangara, Kakembo, & Lim, 2012; Millward & Mersey, 1999; Perovic et al., 2013; Wang, Yu, Shrestha, Ishidaira, & Takeuchi, 2010; Zhu, 2015). USLE has been widely applied at watershed and catchment scales (Baban & Yusof, 2001; Dickinson & Collins, 1998; Jain & Kothyari, 2000; Jain, Kumar, & Varghese, 2001) on the basis of lumped approach using RS and GIS. However, appropriate watershed management plans for sustainable development require prioritization of watersheds on the basis of micro-units, which contribute to maximum soil erosion. In this context, the past studies (Chowdary, Yatindranath, Kar & Adiga, 2004; Khan, Gupta & Moharana, 2001; Saxena, Verma, Chary, Srivastava & Barthwal, 2000; Sharda, Kumar, Venkataratnam & Rao, 1993; Welde, 2016; Yoshino & Ishioka, 2005) show that proper integration of USLE with RS and GIS techniques can be helpful for prioritization of erosion prone areas. It is also found from the past studies that integration of the USLE model with GIS on grid-cell basis would allow the analysis of spatially distributed soil erosion effectively (Bhattarai & Dutta, 2007; Onori, De Bonis, & Grauso, 2006; Onyando, Kisoyan, & Chemelil, 2005; Renschler, Diekkrüger, & Mannaerts, 1999). Recently, grid-cell based studies have been conducted to identify vulnerable areas in the watersheds and large catchments for the planning of conservation practices (Dabral, Baithuri, & Pandey, 2008; Pandey, Chowdary, & Mal, 2007; Wang et al., 2010, Perovic et al., 2013; Farhan & Nawaiseh, 2015).

USLE estimates the soil erosion caused by a rainfall event, but does not provide sediment yield at the outlet of the watershed. Vanoni (1975) reported the bulk of sediment deposits in-between sites, wherever the runoff water is inadequate to transport eroded soil to the watershed outlet. Thus, some portion of the eroded soils routing towards watershed outlet are responsible for the sediment yield. Hence, sediment delivery ratios (SDR) need to be determined for adjustment of soil erosion rate estimated by USLE to quantify the sediment yield at the watershed outlet. A reliable assessment of SDR is done by using observed sediment yield at watershed outlet and estimated soil erosion by USLE over the watershed (Ambika, Satoshi, & Okihiro, 2006; Weifeng & Bingfang, 2008). The modification was made on USLE in the form of MUSLE to estimate sediment yield more effectively under different conditions (Lal, 2001). Although, MUSLE performs better than USLE in estimating sediment yield, it does not provide an appropriate assessment of spatially distributed soil erosion (Wang, Hapuarachchi, Ishidaira, Kiem, & Takeuchi, 2009).

In the last decade, good amount of work has been done on soil erosion assessment in large catchments and at a regional scale by using lumped approaches and very few on micro-units and grid-cell basis. However, estimation of spatially distributed soil erosion on grid-cell basis has not been adequately addressed, in the agricultural watersheds, which has effect on fertility of agricultural land and quality of water. Therefore, the reported study is carried out with the following specific objectives: (i) Estimation of average annual soil erosion on 100 m × 100 m grid-cells basis by integrating the USLE model with GIS in a small agricultural watershed (ii) Classification of the estimated average annual soil erosion on grid-cell basis into different soil erosion classes (iii) Identification of the critical sub-watershed and critical erosion prone grid-cells that require urgent conservation measures and land management (iv) Estimation of sediment delivery ratio (SDR) for each hydrological unit to provide the proportion of the eroded soil reaching the outlet.

2. Material and methods

2.1. Study area

A small agricultural watershed namely Kapgari watershed (KGW), located in the eastern India was selected for the study. Geographically the watershed lies between 86°50' and 86°55'E longitude and 22°30' and 22°35' N latitude (Fig. 1) and has an area of 973 ha. The watershed was delineated into three sub-watersheds (Fig. 1) on the basis of the main drain and three sub-drains. The sizes of delineated sub-watersheds KGSW1, KGSW2 and KGSW3 are 280, 330 and 363 ha respectively. The climate of the study area is sub-humid subtropical and the average annual rainfall is 1370 mm. About 80% of the annual rainfall is concentrated during the rainy season from June to October. The daily average temperature varies from 24 °C to 40 °C and that of relative humidity from 59.4% to 94.3%. The major soil textural classes of the study watershed are sandy loam, silt loam, clay loam and loam; sandy loam soil being predominant. Erosion is the major problem of this agricultural watershed due to its undulating topography and unmanaged natural resources.

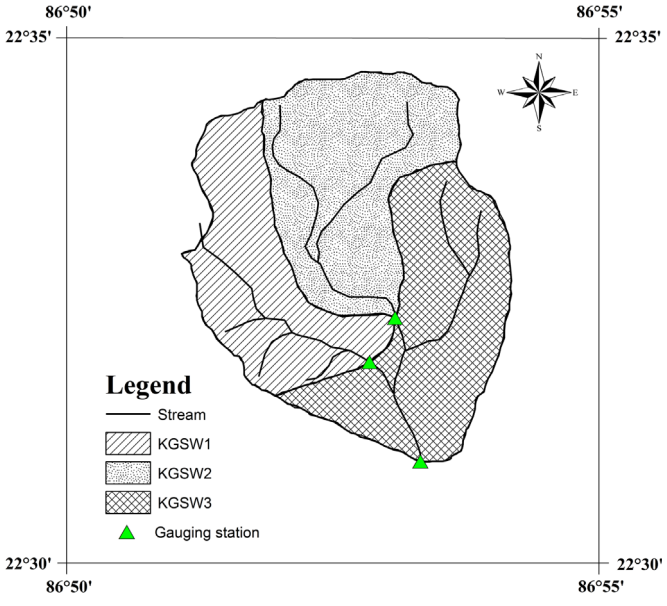


Fig. 1. Location map of Kapgari watershed.

2.2. Data acquisition

2.2.1. Hydro-metrological data

The daily rainfall as well as rainfall intensity during the monsoon season was collected for three years (2003–2005) from the metrological observatory setup in the watershed. The sediment yield from the whole watershed and its sub-watersheds were measured for the years: 2003, 2004 and 2005 through manual sampling at the outlet after the storm (Singh & Panda, 2015). The water samples for each storm were collected from the channel using bottle sampler at one hour intervals. Main outlet of the watershed and the outlets of sub-watersheds KGSW1 and KGSW2 were gauged for effective monitoring of surface runoff (Singh, Panda, & Lamers, 2015). Separate measurement for KGSW3 was not necessary, since the main outlet of watershed and outlet of KGSW3 is the same (Fig. 1). Current meters were utilized to measure the flow velocity, and automatic stage level recorders were used to record the fluctuations of water level. Water samples collected from the outlet of watershed and its sub-watersheds were passed through filter papers (Watman No.1). The suspended sediment retained on the filter paper was dried and weighed. The suspended sediment yield (SY) for each day was computed using the general formula:

$$SY = 3600 \sum_{i=1}^{n} Q_i C_i \quad (1)$$

where, SY is the sediment yield (kg); Q_i is the observed stream discharge ($m^3\ s^{-1}$) at time i; C_i is the observed sediment concentration at time i ($g\ l^{-1}$) and n is the duration of the event (h).

2.2.2. Remote sensing data

Nearly zero percent cloud cover remote sensing data of Indian Remote Sensing Satellite (IRS-1D LISS- III) of Row no-107 and Path no-56 for 14th November 2003 were collected from National Remote Sensing Center (NRSC), Hyderabad, India. The spatial resolution of the LISS-III satellite data is 23.5 m with a ground swath of 141 km. The digital image was rectified and geometrically corrected by transferring coordinates of the identifiable features such as: road crossing, canal, bridges etc. from geometrically corrected topographical sheet to satellite image.

2.3. Universal soil loss equation (USLE)

The USLE is an erosion prediction model, which predicts only the losses from sheet and rill erosion under specified cropping and management system condition. The USLE model is suitable for estimating long-term average soil erosion using following empirical equation (Wischmeier & Smith, 1978):

$$A = R \times K \times L \times S \times C \times P \quad (2)$$

where, A is average annual soil erosion per unit area ($t\ ha^{-1}$); R is rainfall erosivity factor ($MJ\ mm\ ha^{-1}\ h^{-1}$); K is the soil erodibility factor ($Mg\ h\ MJ^{-1}\ mm^{-1}$); LS is the topographic factor (L represents slope length factor and S represents slope steepness factor); C is the crop management factor and P is the conservation practice factor.

2.4. Sediment delivery ratio (SDR)

Sediment delivery ratio (SDR) is a fraction of the eroded soil from the source area transporting to the sink area with surface flow. Usually, total estimated soil erosion by USLE is considerably higher than the observed sediment yield, due to deposition of the eroded soil at intermediate locations of the watershed (Vanoni, 1975). SDR can be used to adjust the soil erosion rate estimated by USLE for estimating sediment yield at the outlet. Eq. (3) is a mathematical formulation for estimating SDR, where Y represents the observed sediment yield at the outlet of the watershed and E represents the estimated average annual soil erosion using USLE for the same watershed (Walling, 1983).

$$SDR = \frac{Y}{E} \quad (3)$$

3. Methodology

In the present study, the USLE model was integrated with GIS on grid-cell basis for analysis of soil erosion in detail. Based on the research findings of Renschler et al. (1999), a grid-cell size of 100 m × 100 m was selected for estimation of the soil erosion. The thematic layers of USLE models were generated in GIS environment to estimate the spatial distribution of the soil erosion on 100 m × 100 m grid scale. It was assumed that each grid-cell is a closed plot of one hectare, and has no interaction with the other grid-cell. The estimated soil erosion by the USLE model for different rainfall events and the measured sediment yield were used to estimate the SDR to determine the proportion of the eroded soil reaching the outlet. The approach used to estimate soil erosion and SDR for the present study is shown in Fig. 2 and the methodology followed is explained in the following sections.

3.1. Delineation of watershed parameters

The Survey of India (SOI) toposheet number: 73/ J-14 in the scale of 1:50,000 were utilized for extracting the topographical features of the study watershed. The hard copy of SOI topographic sheet was scanned and then the scanned image was rectified and geometrically corrected. The contour lines on SOI toposheet with 10 m interval were digitized in arc/line mode using 'Sketch tool' in a GIS platform (ESRI ArcGIS 10.2). The digitized contours were interpolated to develop a Digital Elevation Model (DEM) for the watershed. Further, the DEM was used to delineate watershed boundary and drainage network through 'Hydrology tool' in the GIS platform. The DEM of the study watershed is presented in the Fig. 3, which shows that the maximum elevation difference is 81 m.

The present study is based on 100 m × 100 m grid-cells as operational unit for estimation of the soil erosion. Thus, the polygon watershed boundary was converted into grid and the resolution of grids of 100 m x 100 m size. Each grid-cell was represented by a distinctive number and the numbering of grids was done starting from northwest corner and then going right and then down.

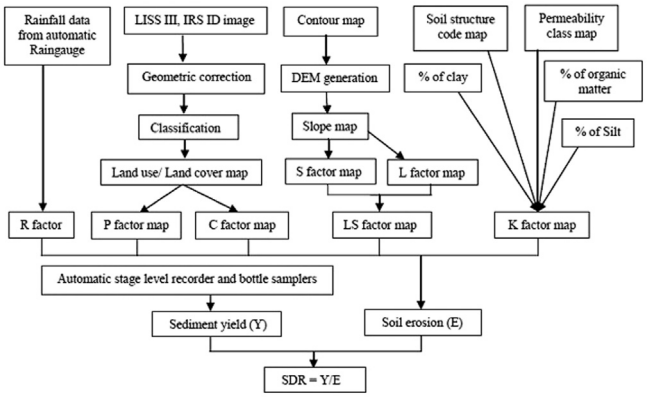


Fig. 2. Schematic diagram for assessment of soil erosion and sediment delivery ratio (SDR).

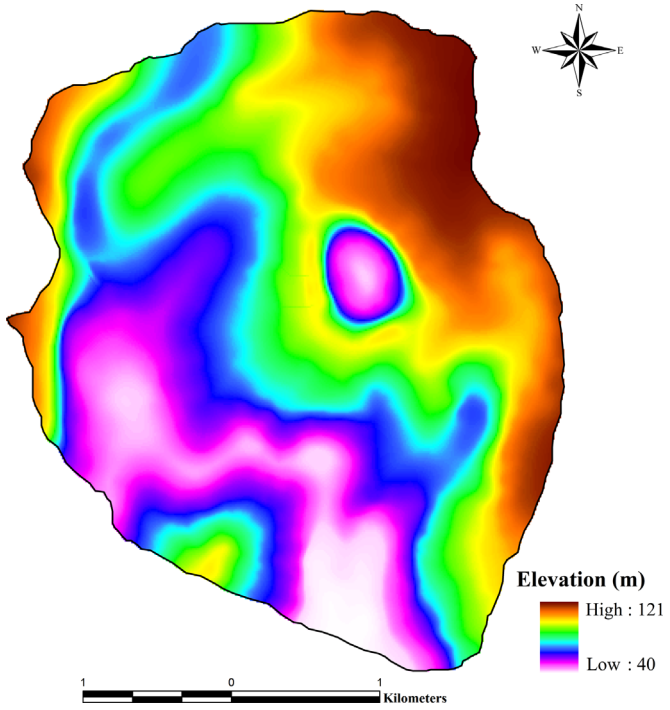


Fig. 3. Digital Elevation Model of Kapgari watershed.

3.2. Development of database for USLE

3.2.1. Rainfall erosivity factor (R)

In the reported study, Rainfall erosivity factor (R) was estimated using daily rainfall magnitude as well as intensity data for a period of three years (2003–2005). The spatial distribution of R factor was suppose to be homogeneous due to the small size of the watershed (973 ha). The R factor was estimated as the product of the total storm kinetic energy (E) and the maximum 30-min rainfall intensity (I_{30}) occurring during heavy rain, and recognized as the rainfall erosivity index (EI_{30}). The rainfall intensity data were used to compute the I_{30} in any 30-min duration throughout the heavy rain. The storms with rainfall values of 12.5 mm or more and alienated from another storm by at least 6-h were opted for computing rainfall erosion index (Satpathi, Panda, & Prasad, 1999). The kinetic energy was estimated for the rainfall periods having constant intensity, using the following equation (Wischmeier & Smith, 1978).

$$E_i = P_i(0.119 + 0.0873\log_{10} I_i) \text{ for } I_i < 76 \text{ or} \quad (4a)$$

$$E_i = P_i \times 0.283 \text{ for } I_i > 76 \quad (4b)$$

where, E_i is the kinetic energy (MJ ha^{-1}); P_i is the depth of rainfall (mm) and I_i is the intensity of rainfall (mm h^{-1}) for a rainfall periods having constant intensity.

Finally, the R factor for n number of periods was calculated using total kinetic energy of a storm (E) for k number of such type of periods using the following relationships.

$$E = \sum_{i=1}^{k} E_i \quad (5)$$

$$EI_{30} = E \times I_{30} \quad (6)$$

$$R = \sum_{l=1}^{n}\left[\sum_{j=1}^{m}(EI_{30})_j\right] \quad (7)$$

where, I_{30} is the highest rainfall intensity in any 30 min duration (mm h^{-1}); EI_{30} is the rainfall erosivity index for storm j; m is the number of storms in n number of periods; R is rainfall erosivity factor (MJ mm ha^{-1} h^{-1}) for n number of periods.

3.2.2. Soil erodibility factor (K)

The soil erodibility factor (K) is the rate of soil erosion per unit of rainfall erosivity index for a specified soil. The K factor was calculated using the following regression equation (Foster, McCool, Renard, & Moldenhauer, 1981).

$$K = 2.8 \times 10^{-7} \times M^{1.14}(12 - a) + 4.3 \times 10^{-3}(b - 2) + 3.3 \times 10^{-3}(c - 3) \quad (8)$$

where, K = soil erodibility factor (Mg h MJ^{-1} mm^{-1}); M = particle size parameter (percent silt + percent very fine sand) (100 – percent clay); a = percent organic matter content; b = soil structure code; c = soil profile permeability class.

For estimation of the K factor, the soil texture map was collected from the National Bureau of Soil Survey and Land Use Planning (NBSS&LUP) and scanned, rectified and geometrically corrected. Further, the soil textured polygons (vector format) were extracted from soil texture map using digitized boundaries of all villages of Kapagari watershed. The undisturbed soil samples (0–15 cm layers) were collected from each soil texture polygon of the watershed for estimation of the soil properties. The locations of the collected soil samples are shown in the Fig. 4 and estimated soil properties of the collected samples are presented in Table 1. Different thematic layers of soil properties such as percentage of sand, silt, clay, organic matter, soils structure code and permeability classes were generated from ground truth information collected from study area. Soil structure and permeability code were provided based on the particle size and permeability rate of soil, as presented in Tables 2 and 3 respectively. These thematic maps were coded into the module which was developed in ModelBuilder of the GIS platform to generate the K factor thematic layer using the regression Eq. (8). The K factor layer, which was in raster format was changed to vector format and intersected with the watershed grid coverage to find out 100 m × 100 m grid-cell wise soil properties of the study watershed.

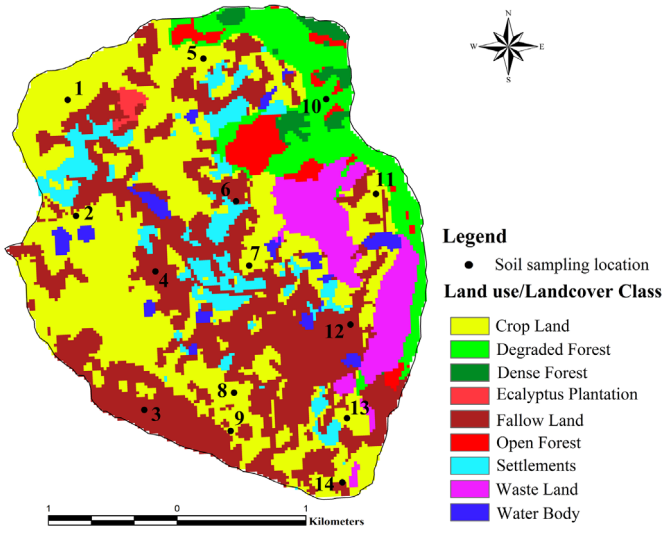


Fig. 4. Land use/ Land cover classification of Kapgari watershed along with the sampling locations.

Table 1
Soil properties of Kapgari watershed.

Sl. No.	Latitude	Longitude	Soil Texture	Soil fraction (%)			Organic matter (%)	Permeability rate mm/h
				Sand	Silt	Clay		
1	22.53	86.86	Sandy Loam	79.23	15.38	5.39	0.95	12.26
2	22.53	86.87	Loamy Sand	80.85	11.58	7.57	0.84	13.45
3	22.51	86.87	Sandy Loam	75.58	14.59	9.83	0.99	9.83
4	22.52	86.87	Sandy Loam	76.45	13.16	10.39	0.92	10.95
5	22.54	86.88	Sandy Loam	70.62	24.59	4.79	1.17	11.53
6	22.53	86.88	Sandy Loam	70.02	24.59	5.39	1.08	11.32
7	22.52	86.88	Loamy Sand	81.44	13.98	4.58	0.86	13.68
8	22.51	86.88	Clay Loam	41.52	37.99	20.49	1.29	1.28
9	22.51	86.88	Clay Loam	41.62	37.19	21.19	1.19	1.31
10	22.53	86.89	Sand	81.02	14.79	4.19	0.91	14.20
11	22.53	86.89	Sandy Loam	76.03	13.68	10.29	1.03	9.46
12	22.52	86.89	Loam	48.22	38.79	12.99	1.17	2.74
13	22.51	86.89	Loam	48.62	38.19	13.19	1.20	3.36
14	22.51	86.89	Loam	47.78	36.41	15.81	1.12	3.58

Table 2
Structure code for different types of soil.

Code	Structure	Particle size (mm)
1	Very fine granular	<1
2	Fine granular	1–2
3	Medium or coarse granular	2–10
4	Blocky, platy or massive	>10

Table 3
Permeability code for different types of soil.

Code	Description	Permeability rate (mm/h)
1	Rapid	>130
2	Moderate to rapid	60–130
3	Moderate	20–60
4	Slow to moderate	5–20
5	Slow	1–5
6	Very slow	<1

3.2.3. Topographic factor (LS)

The topography factors such as: slope length (L) and slope steepness (S) were calculated using the following equations.

(a) L factor: It was calculated based on the relationship developed by (Wischmeier & Smith, 1978).

$$L = \left(\frac{\lambda}{22.13}\right)^{m} \quad (9)$$

where, λ is field slope length (m); m is the dimensionless exponent that depends on slope, being 0.5 if slope $>5\%$, 0.4 if slope $\leq 5\%$ and $>3\%$, 0.3 if slope $\leq 3\%$ and $>1\%$, 0.2 if slope $\leq 1\%$.

In the reported study the DEM of the study area was interpolated for 100 m × 100 m grids using resampling technique to derive the slope map of the watershed at 100 m × 100 m scale. The DEM derived slope map was utilized to develop slope length (L) map using grid size of 100 m as field slope length (λ), based on the recommendation made by previous researchers (Fistikoglu & Harmancioglu, 2002; Jain et al., 2001; Pandey, Mathur, Mishra, & Mal, 2009).

(b) (b) S factor: It was estimated using the following equation (McCool, Brown, Foster, Mutchler, & Meyer, 1987), having length of the slope higher than 4 m.

$$S = 10.8 \sin\theta + 0.03 \quad for\ slope < 9\% \quad (10a)$$

$$S = 16.8 \sin\theta - 0.05 \quad for\ slope \geq 9\% \quad (10b)$$

where, θ represents angle of the slope (degrees).

A single topographic factor (LS) was estimated with a module developed in ModelBuilder of the GIS platform using the Eqs. (9) and 10.

3.2.4. Crop management factor (C) and Conservation practice factor (P)

The land use/land cover (LU/LC) characteristics of an area is the basis of describing the C and P factors. Thus, LU/LC classification of the study watershed was performed using remote sensing data (IRS-1D LISS-III) for 14th November 2003. The LU/LC classification was done using the supervised classification method, in which the operator classifies an area or group of pixels that belong to one or more categories of specific LU/LC. The maximum likelihood classifier (MLC) as described by Lillesand, Kiefer, and Chipman (2014), quantitatively evaluates both the variance and co-variance of the category training data, which is normally distributed (Gaussian), was used during classification process. LU/LC map of the year 2003 was also used for the year 2004 and 2005 due to minor changes in the cultivation practices in this small agricultural watershed. The LU/LC map was resampled over 100 m × 100 m grid-cell using Nearest-neighbour interpolation method. Further, LU/LC map was intersected with the watershed grid coverage to find out 100 m × 100 m grid-cell wise land use /land cover values. Then this grid-cell wise LU/LC was used to assign the C and P factors for diverse LU/LC classes of the study watershed. The C and P values were chosen based on the research findings of Singh, Babu, and Chandra (1981) as well as by Rao (1981), Singh, Babu, Narain, Bhushan, and Abrol (1992) and USDA Soil Conservation Service (1972) handbook.

4. Results and discussion

The USLE factors were estimated on 100 m × 100 m grid scale to provide the spatial distribution of average annual soil erosion in the Kapgari watershed. The soil erosion from Kapgari watershed and its sub-watersheds were also estimated separately for each year and compared with the measured values of sediment yield.

Table 4
Rainfall erosivity factor (R) of Kapgari watershed.

Year	$(EI_{30})_{min}$ (MJ mm ha^{-1} h^{-1})	$(EI_{30})_{max}$ (MJ mm ha^{-1} h^{-1})	R_{Annual} (MJ mm ha^{-1} h^{-1})
2003	60.47	680.88	6286.17
2004	9.48	1268.41	7248.41
2005	17.19	1349.31	4998.88
Average annual R factor = 6177.82 MJ mm ha^{-1} h^{-1} yr^{-1}			

Further, these annual soil erosion values were used to estimate the yearly SDR for each hydrological unit to provide the proportion of the eroded soil reaching the outlet.

4.1. Estimation of USLE factors

Based on the daily rainfall and rainfall intensity data, storm erosivity index (EI_{30}) and annual rainfall erosivity factor (R) were estimated for the years 2003, 2004 and 2005, and are presented in Table 4. It was observed that the annual R factor is the lowest for the year 2005 but has the highest EI_{30} for some storms, which is due to presence of less number of rainfall events with high intensity during this year as compared to the years 2003 and 2004. The average annual R factor calculated for Kapgari watershed was 6177.82 MJ mm ha^{-1} h^{-1} yr^{-1}. The grid wise spatial distribution and magnitude of soil erodibility factor (K) is presented in Fig. 5, which varies between 0.0165 Mg h MJ^{-1} mm^{-1} and 0.0385 Mg h MJ^{-1} mm^{-1}. Low values represent more resistance of the soil against detachment by high intensity rainfall, because the soil of this area has a fine structure, with clay and high organic matter content. The mean soil erodibility factor for Kapgari watershed was 0.0256 Mg h MJ^{-1} mm^{-1}. The magnitude and spatial distribution of LS factor is shown in Fig. 6. A major portion of the watershed has a LS factor of less than 0.70; the ranges of LS factor vary between 0.051 and 2.0 and the mean LS factor is 0.643.

A total of nine LU/LC classes were derived from satellite imagery and were used to assign magnitude and spatial distribution of the C and P factors. The supervised classification along with MLC algorithm was used for the LU/LC classification, and its accuracy assessment (error matrix) was done using ground truth data collected from the survey of India toposheet and field data collected from the study watershed. The error matrix (Table 5) was

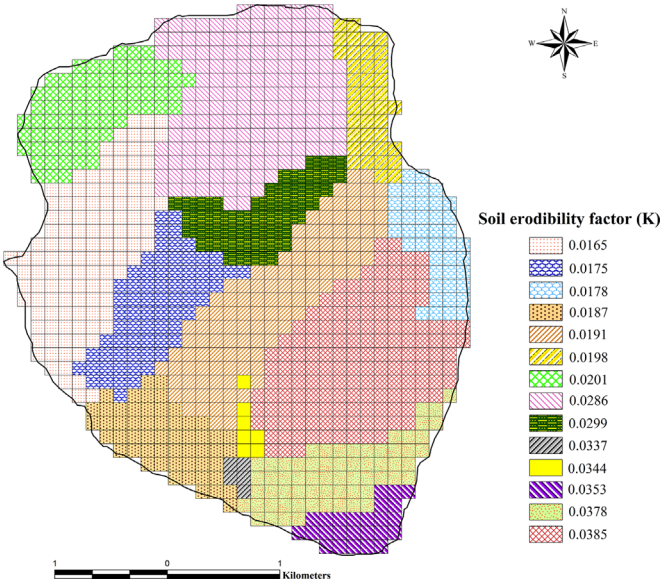


Fig. 5. Spatial distribution of the soil erodibility factor (K) in Kapgari watershed.

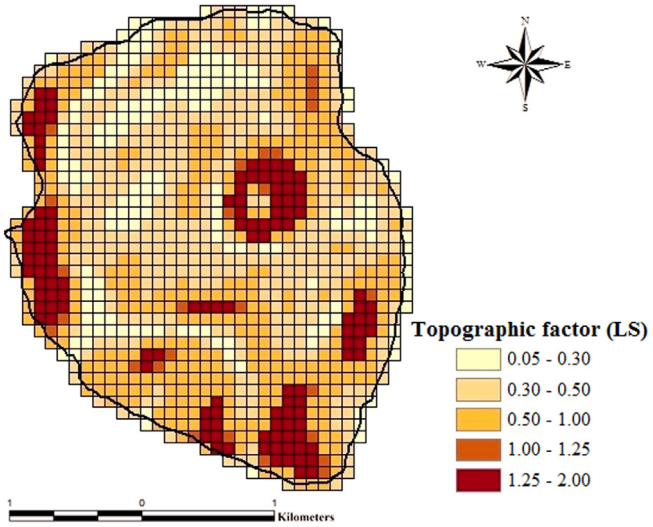


Fig. 6. Spatial distribution of the topographical factor (LS) in Kapgari watershed.

estimated using 297 random samples which represent different LU/LC classes. Overall accuracy of the classification was found to be 83.16% and omission and commission errors are presented in Table 5. In addition, the Kappa coefficient (K_{hat}) developed by Cohen (1960) was estimated to evaluate observer agreement for categorical data. The estimated K_{hat} (0.802) represents strong agreement and good accuracy based upon the criteria given by Manserud and Leemans (1992). The classified image representing various LU/LC classes of the study area is shown in Fig. 4 and its statistics are presented in Table 6. The area classified under wasteland represents the degraded land having coarse textured soil which is slightly acidic in nature containing low level of N, P and K. The wasteland is underutilized and has a gentle slope without any conservation practices. Finally, C and P factors were assigned different land use patterns as given in Table 6. The magnitude and spatial distribution of C and P factor are given in Figs. 7 and 8 respectively. The C factor varies from 0 to 0.5 and that of P factor from 0 to 1.

4.2. Annual soil erosion assessment by USLE

The annual soil erosion from the watershed was estimated as the product of annual rainfall erosivity factor and 100 m × 100 m grid-cell based thematic layers of K, LS, C and P factors. Subsequently, annual soil erosion for each sub-watershed was computed as the total annual soil erosion of the sub-watershed divided by the total geographical area of that sub-watershed. The estimated soil erosion from Kapgari watershed and its sub-watersheds (KGSW1, KGSW2 and KGSW3) for each year has been presented in Table 7. Wide variation was found in soil erosion rate, which is mainly due to the variation in R factor. The estimated soil erosion rate was compared with the measured sediment yield at the outlet to validate the USLE model. The daily soil erosion was estimated, using daily rainfall erosivity index (EI_{30}) and plotted against the daily observed sediment yield at each gauging site. The scatter plot (Fig. 9) of estimated soil erosion and observed sediment yield for the whole watershed and its sub-watersheds show mostly over-prediction, even though the model predicts even sediment yield of small magnitude with considerable accuracy. The overestimation of the soil erosion may be due to the fact that the USLE model estimates only the soil losses through sheet and rill erosion under specified cropping and management conditions and does not provide the proportion of the eroded soil (sediment yield) reaching the outlet. The eroded soil may be deposited at intermediate

Table 5
Accuracy assessment of Land use/Land cover classification.

Classification data	Training datasets										
	CL	DF	DnF	EP	FL	OF	S	WL	W	Row Total	CE
CL	62	3	0	0	0	0	0	2	3	70	11.43
DF	2	26	3	0	0	0	0	1	0	32	18.75
DnF	0	2	15	0	0	1	0	0	0	18	16.67
EP	0	0	0	5	0	2	0	0	0	7	28.57
FL	0	0	0	0	57	0	3	4	0	64	12.50
OF	3	1	2	1	0	21	0	0	0	28	25.00
S	0	1	0	0	3	0	19	2	0	25	24.00
WL	2	0	0	0	4	0	2	28	0	36	22.22
W	0	2	0	0	0	0	1	0	14	17	17.65
Column Total	69	35	20	6	64	24	25	37	17	**297**	
OE	10.14	25.71	25.00	16.67	10.94	12.50	24.00	24.32	17.65		

CL: Cropland; DF: Degraded forest; DnF: Dense forest; E: Ecalyptus plantation; FL: Fallow land, OF: Open forest; S: Settlement; WL: Wasteland; W: Water body; CE: Commission error; EO: Omission error

Table 6
Analysis of Land use/Land cover of Kapgari watershed and typical values of C and P factor for different Land use/Land cover classes.

Land use/Land cover classes	% Area				C factor	P factor
	KGSW1	KGSW2	KGSW3	Kapgari		
Crop land (Rice)	59.71	32.30	23.47	36.89	0.28	0.03
Degraded forest	0.00	13.53	12.23	9.15	0.13	0.90
Open forest	0.00	4.26	4.05	2.96	0.02	0.85
Dense forest	0.00	2.43	2.08	1.60	0.01	0.80
Plantation	0.00	1.39	0.00	0.47	0.13	0.80
Fallow land	33.02	31.32	35.32	33.30	0.28	0.03
Settlement	4.95	12.24	2.94	6.67	0.50	0.10
Waste land	0.00	0.23	17.72	6.69	0.14	1.00
Water bodies	2.32	2.30	2.19	2.27	0.00	0.00

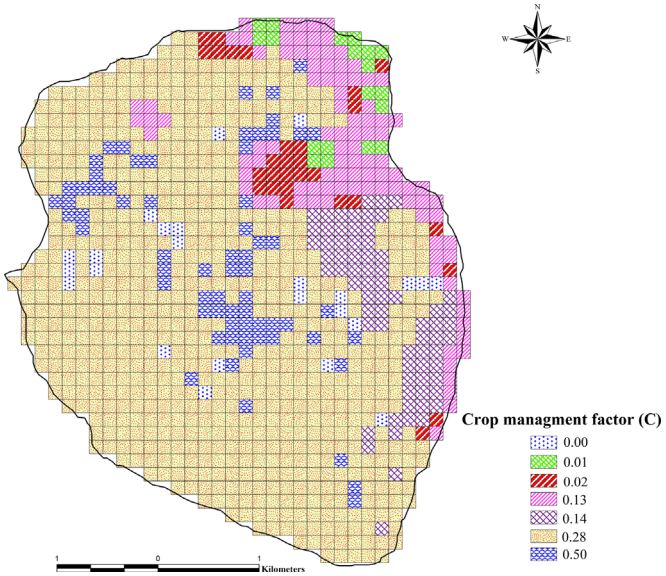


Fig. 7. Spatial distribution of the crop management factor (C) in Kapgari watershed.

locations (e.g. rice crop check basins) of the watershed, in cases wherever the runoff water is inadequate to transport eroded soil to the outlet. In spite of that, it can be seen from Fig. 9 that coefficient of determinations (R^2) between estimated soil erosion and observed sediment yield were 0.71, 0.77 and 0.64 for whole watershed and its sub watershed KGSW1 and KGSW2 respectively.

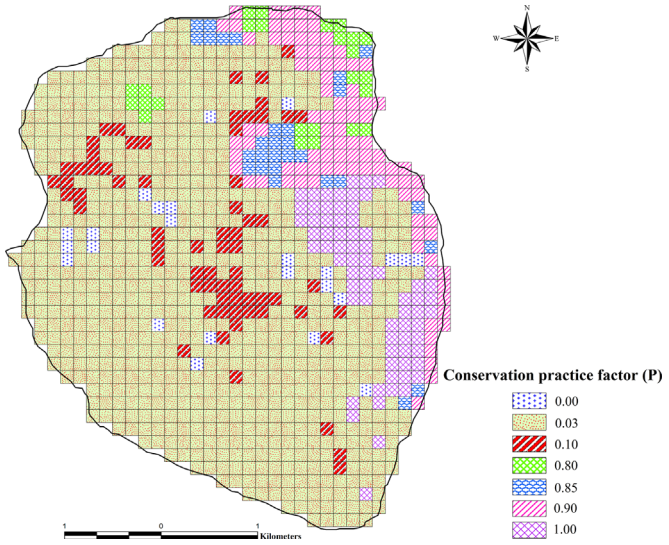


Fig. 8. Spatial distribution of the conservation practice factor (P) in Kapgari watershed.

Table 7
Annual soil erosion (t ha^{-1}) using USLE model in Kapgari watershed.

Year	Kapgari	KGSW1	KGSW2	KGSW3
2003	3.66	0.80	2.74	**6.98**
2004	4.22	0.93	2.93	**8.02**
2005	2.91	0.64	2.02	**5.53**

These R^2 values indicate a good relationship between the estimated soil erosion and observed sediment yield. It was also observed that the KGSW1 has the R^2 value higher than that of the KGSW2.

4.3. Estimation of average annual soil erosion using the USLE

In the present study, 3-years averaged R factor of 6177.82 MJ mm ha^{-1} h^{-1} yr^{-1} was used to estimate average annual soil erosion from the Kapgari watershed. The average annual soil erosion in the watershed was estimated as the product of USLE parameters at 100 m × 100 m grid scale. Finally, the estimated average annual soil erosion on a grid-cell basis was regrouped and classified into the priority scales such as: "Slight (0–5 t ha^{-1} yr^{-1}), Moderate (5–10 t ha^{-1} yr^{-1}), High (10–20 t ha^{-1} yr^{-1}), Very high (20–40 t ha^{-1} yr^{-1}), Severe (40–80 t ha^{-1} yr^{-1}) and Very severe

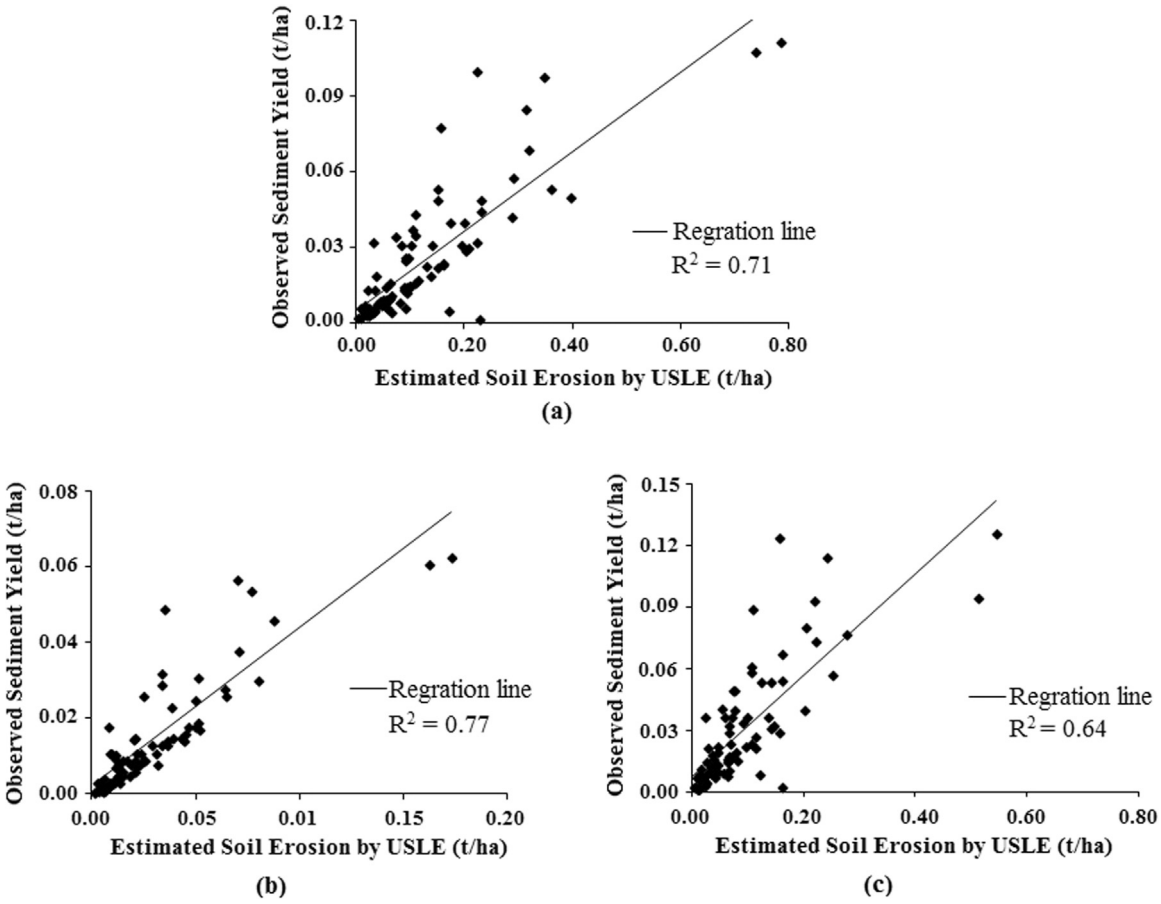


Fig. 9. Relationship between observed daily sediment yield at outlet and estimated soil erosion using USLE (a) Kapgari watershed (b) KGSW – 1 (c) KGSW-2.

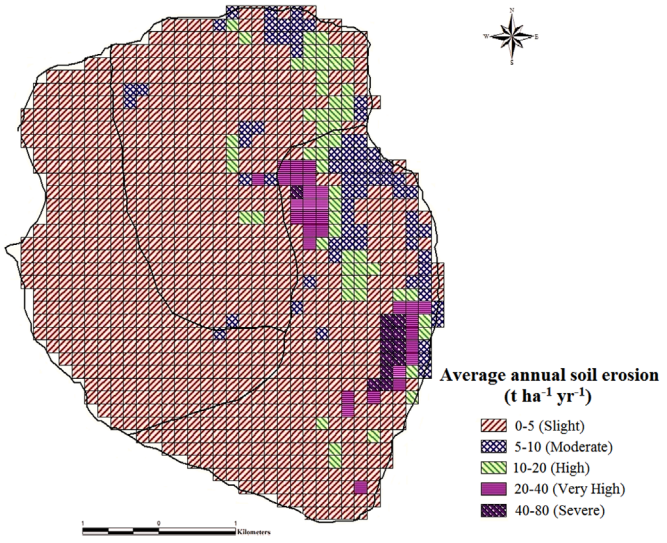


Fig. 10. Spatial distribution of average annual soil loss of Kapgari watershed.

(> 80 t ha^{-1} yr^{-1}) erosion classes as per the guidelines suggested by Singh et al. (1992) for Indian conditions" (Pandey et al., 2007, p.742). The spatial distribution of average annual soil erosion from Kapgari watershed on 100 m × 100 m grid-cells is presented under Fig. 10. The area under different soil erosion severity classes are also presented in Table 8. The average annual soil erosion from the whole watershed and its sub-watersheds KGSW1, KGSW2 and KGSW3 were also estimated and found to be 3.68, 0.81, 2.62 and 7.01 t ha^{-1}yr^{-1} respectively.

Table 8
Statistics of different soil erosion classes in Kapgari watershed.

Soil erosion (t ha^{-1} yr^{-1})	Area (percent)	Soil erosion class
0–5	82.63	Slight
5–10	**6.87**	Moderate
10–20	**5.96**	High
20–40	**3.30**	Very high
40–80	**1.24**	Severe
> 80	0.00	Very severe

The results clearly indicate that the average soil erosion from KGSW1 contribute very little to the total soil erosion and has the magnitude of only 0.81 t ha^{-1} yr^{-1}. It was also observed from Fig. 10 that all the grids-cells of KGSW1 comes under slight class of soil erosion except one grids-cell which is in the moderate class of soil erosion. The grids having slight class of soil erosion under KGSW1 are predominated by rice fields, where rice check basins provide good conservation practices. The KGSW2 contributed to soil erosion significantly (2.62 t ha^{-1} yr^{-1}), since degraded, open forest and dense forest area are present in this sub-watershed, where conservation practices are very poor. The KGSW2 was having significant number of moderate and high class of soil erosion grids except one grid-cell which comes under very high class of soil erosion (Fig. 10). The grid having very high class of soil erosion has open forest with higher LS and K factor values of 1.25 and 0.029 Mg h MJ^{-1} mm^{-1} respectively (Figs. 4–6). It was found that the most critical erosion prone area is the KGSW3 having the highest soil erosion (7.01 t ha^{-1} yr^{-1}) due to relatively high surface slope, undulating topography and the highest area under wasteland (17.72%). The KGSW3 has more number of moderate,

Table 9
Calculation of SDR for Kapgari watershed and its sub-watersheds.

Unit Name	SY 2003	SE 2003	SY 2004	SE 2004	SY 2005	SE 2005	SDR 2003	SDR 2004	SDR 2005	SDR Average
KGSW1	0.37	0.81	0.36	0.93	0.43	0.64	0.46	0.39	0.67	**0.51**
KGSW2	0.95	2.54	0.66	2.93	0.84	2.02	0.37	0.23	0.41	**0.34**
Kapgari	0.73	3.66	0.66	4.22	0.72	2.91	0.20	0.16	0.25	**0.20**

SY, SE, SDR represent sediment yield, soil erosion and sediment delivery ratio

high and very high classes of soil erosion grids as compare to other sub-watersheds (Fig. 10). This critical sub-watershed has 12 grid-cells under severe class of soil erosion, which has wasteland with highest LS and K factor values of 2.00 and 0.038 Mg h MJ^{-1} mm^{-1} respectively (Figs. 3–5), and might have contributed to the highest soil erosion.

The results illustrate that 82.63% area of the watershed is classified under slight erosion class, 6.87% under moderate, 5.96% under high, 3.30% under very high; whereas only 1.24% is classified under "severe erosion class" and none of the grid-cells falls under “very severe erosion class" (Table 8). However, in near future there are chances that the soil erosion within the study area will be increased due to utilization of more area under cultivation by cutting of trees to meet the food demand of the ever growing population. For this reason, the Kapgari watershed requires immediate attention for the grid-cells having moderate to "severe erosion class" (approximately 17.37% area) from the soil conservation point of view.

4.4. Estimation of sediment delivery ratio

The USLE model cannot simulate either the gully erosion or the erosion in the channels, because this model gives a clear indication of the amount of soil eroded every year in the watershed and not how much eroded soil really reached the outlet. Thus, in the present study SDR was estimated for each hydrological unit to provide proportion of the eroded soil reaching the outlet. The SDR was calculated using observed sediment yield data for the years 2003, 2004 and 2005 and are presented in Table 9. The results show that KGSW1 and KGSW2 have average SDR of 0.51 and 0.34, which means that 49% and 66% of eroded soil gets deposited at the intermediate locations before reaching the outlets respectively. It was also observed that the average SDR estimated for the whole watershed was 0.20, which means that 80% of soil gets deposited in-between locations of the watershed, and only 20% of the eroded soil reaches the outlet.

The results revealed that soil erosion from KGSW3 is the highest (7.01 t $ha^{-1}yr^{-1}$) followed by KGSW2 (2.62 t $ha^{-1}yr^{-1}$) and KGSW1 (0.81 t $ha^{-1}yr^{-1}$). However, SDR estimated at the outlet of whole watershed is lowest (0.20), whereas SDR of KGSW1 and KGSW2 are 0.51 and 0.34 respectively. Therefore, it may be inferred that even if more soil erosion is obtained from KGSW3, mostly it gets deposited before reaching the outlet of the watershed. The reason being that eroded soils from the critical erosion prone areas of KGSW3 are trapped in rice crop fields, fallow land and water bodies before reaching the outlet, since these LU/LC are just below the critical erosion prone areas (Figs. 4 and 9). These inferences are just assumptions based on results and visualizations, because values of SDR for an area is influenced by many factors such as, vegetation, catchment physiography, transport system, texture of eroded material and their complex interactions on the land surface (Richard, 1993; Walling, 1983, 1988). The consequences of these interactions make it difficult to classify the principal controls on watershed sediment response and on sub-watershed to sub-watershed variability.

Integration of USLE model with GIS and RS data has significant importance for a preliminary assessment of soil losses through sheet and rill erosion. USLE model does not compute deposition on concave slopes, at dense vegetative strips, in terrace channels, and in sediment basins. Furthermore, process-based equations can be used to estimate the sediment transport capability and soil deposition which requires more accurate and detailed ground acquisition data.

5. Conclusions

The following specific conclusions were drawn from the reported study:

1. Soil erosion rate estimated on 100 m × 100 m grid-cell basis using USLE, RS and GIS matches well with observed sediment yield for the study watershed and its sub-watersheds.
2. A major portion (82.63%) of the total area of the watershed was classified under slight erosion class, whereas the rest falls under the moderate to severe erosion class. However corrective measures are to be taken, so that more area of the watershed do not transform into the severe class of erosion.
3. The study revealed that KGSW3 is a critical sub-watershed having the highest number of critical erosion prone grid-cells, and thereby with the highest priority for soil conservation treatment, followed by KGSW2 and KGSW1in that order.
4. Lowest sediment delivery ratio obtained for the whole watershed as compared to KGSW1 and KGSW2 sub-watersheds indicate that most of the eroded soil from KGSW3 is deposited at intermediate locations such as: rice check basins, before reaching the outlet.
5. Spatial distribution of estimated average annual soil loss on 100 m × 100 m grid-cell basis and information on SDR for the study watershed and its sub-watersheds will be useful to develop management scenarios and guidelines for the policy makers to develop and recommend soil erosion prevention and control measures in critical erosion prone areas under similar conditions.

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